AI-Empowered Space-Air-Ground Integrated Networks: A Survey

Anonymous submission

Abstract

Space-Air-Ground Integrated Networks (SAGIN), which combine satellite, aerial, and terrestrial communication layers, are considered a key architecture for future 6G wireless systems because they provide wider coverage and improved connectivity for applications such as disaster recovery, intelligent transportation, and Internet of Things (IoT) services. However, the mobility, heterogeneity, and dynamic nature of SAGIN make these networks difficult to manage using conventional approaches, leading to increasing interest in Artificial Intelligence (AI) techniques such as deep reinforcement learning, federated learning, agentic AI, and edge AI. This paper surveys recent research on AI-enabled SAGIN and reviews how these techniques have been applied to resource allocation, beamforming, trajectory planning, authentication, and network security. Our analysis identifies several common trends across the literature. Most AI-based solutions are validated through simulations rather than real-world deployments, researchers are increasingly adopting decentralized and autonomous network management approaches, and the majority of studies focus on performance optimization while comparatively less attention is given to security. Based on these findings, we discuss current research gaps and identify directions for advancing practical, scalable, and secure AI-enabled SAGIN for future 6G networks.

Introduction

Placeholder introduction text.

Background

Space-Air-Ground Integrated Networks (SAGIN) are communication networks that combine three different layers: space, air, and ground. The space layer includes satellites, the air layer includes platforms such as drones (UAVs) and high-altitude platforms (HAPS), and the ground layer includes cellular base stations, vehicles, IoT devices, and other terrestrial communication infrastructure. By connecting these three layers, SAGIN can provide wider coverage, improve connectivity, and support communication in areas where traditional ground networks may have limited coverage. This integrated architecture is considered an important technology for future 6G networks because it can support applications such as remote sensing, disaster recovery, intelligent transportation, and Internet of Things (IoT) services.

Throughout the literature reviewed in this survey, several recurring AI techniques are used to manage and optimize SAGIN. This section briefly defines each before they are discussed in more detail in the Literature Review.

Deep Reinforcement Learning (DRL): Deep reinforcement learning is an AI technique that combines reinforcement learning with deep neural networks to solve complex decision-making problems. An AI agent learns by interacting with its environment and improving its decisions based on rewards and penalties. In the papers reviewed, DRL was mainly used to optimize UAV trajectory planning, resource allocation, and task offloading while reducing energy consumption and communication delays.

Federated Learning (FL): Federated learning is a machine learning approach where multiple devices train the same AI model without sharing their raw data with a central server. Instead, they only exchange model updates, which helps improve privacy and reduce communication costs. Some of the papers also proposed decentralized federated learning, where satellites communicate directly with each other instead of relying on a central ground station.

Agentic AI: Agentic AI refers to AI systems that can make decisions and perform tasks with minimal human intervention. Instead of only following predefined instructions, these AI agents can monitor network conditions, make decisions, and adapt to changing environments automatically. In future SAGIN and 6G networks, Agentic AI can help improve resource management, routing, and overall network performance.

Edge AI: Edge AI is the use of AI algorithms on devices located close to where data is generated, such as UAVs, satellites, or edge servers, instead of sending all data to a central cloud. This reduces communication delays, lowers bandwidth usage, and allows faster decision-making, which is important for real-time applications in SAGIN.

Literature Review

Placeholder literature review text.

Discussion

The literature reviewed in this survey reveals several recurring patterns and open challenges in applying AI to Space-Air-Ground Integrated Networks (SAGIN).

One pattern noticed across several of the papers is that most of the proposed AI solutions were only evaluated using simulations instead of real-world deployments. Although the simulation results showed improvements in areas such as resource allocation, energy efficiency, communication delay, and network performance, it is still unclear how well these methods would work in real SAGIN environments. Real networks have additional challenges such as satellite mobility, changing communication links, weather conditions, interference, and limited computing resources that are difficult to capture in simulations. This limitation appears in several papers, including the studies on multi-UAV mobile edge computing (**?**), decentralized federated learning for LEO satellite constellations (**?**), and deep reinforcement learning for task offloading and resource allocation (**?**). This suggests that while AI-based approaches show a lot of promise, more real-world testing is needed before they can be widely deployed in future SAGIN systems.

Another pattern noticed is that many of the papers are moving away from centralized network management and instead using more decentralized or autonomous approaches. For example, the DFedSat paper removes the need for a central ground station by allowing satellites to train AI models directly with each other (**?**). The distributed beamforming paper uses multi-agent reinforcement learning so that each airborne platform can make decisions without constantly sharing channel state information (**?**). The Agentic AI paper also follows this trend by proposing intelligent AI agents that can monitor network conditions and make decisions automatically instead of relying on human operators (**?**). Even though these approaches use different AI techniques, they all aim to reduce communication overhead, improve scalability, and make the network more adaptable. However, they also introduce new challenges, such as coordinating many independent agents and ensuring that decentralized systems remain reliable as the network grows.

A third pattern noticed is that most of the papers focus on improving network performance rather than security. Many of the studies use AI techniques such as deep reinforcement learning, multi-agent reinforcement learning, or Agentic AI to optimize resource allocation, beamforming, task offloading, and communication efficiency. In comparison, only a few papers, such as Chou's work on Edge AI-enabled physical layer security (**?**), focus mainly on protecting the network from security threats. The survey paper by Bakambekova et al. also points out that security remains one of the major open challenges in AI-enabled SAGIN (**?**). This suggests that although researchers are making good progress in improving network performance, more work is still needed to develop AI solutions that can provide both high performance and strong security at the same time.

Proposed Idea

Placeholder proposed idea text.

Conclusion

Placeholder conclusion text.

References

Bakambekova, A.; Kouzayha, N.; and Al-Naffouri, T. 2024. On the Interplay of Artificial Intelligence and Space-Air-Ground Integrated Networks: A Survey. *arXiv preprint arXiv:2402.00881*.

Chou, H.; Solanki, S.; Ha, V. N.; Chen, L.; Ma, S. L.; Al-Hraishawi, H.; Eappen, G.; Querol, J.; Merlano-Duncan, J. C.; and Chatzinotas, S. 2024. Edge AI Empowered Physical Layer Security for 6G NTN: Potential Threats and Future Opportunities. *arXiv preprint arXiv:2401.01005*.

Dev, K.; Khowaja, S. A.; Singh, K.; Zeydan, E.; and Debbah, M. 2025. Advanced Architectures Integrated with Agentic AI for Next-Generation Wireless Networks. *arXiv preprint arXiv:2502.01089*.

Huang, C.; et al. 2024. Joint Offloading and Resource Allocation for Hybrid Cloud and Edge Computing in SAGINs: A Decision Assisted Hybrid Action Space Deep Reinforcement Learning Approach. *arXiv preprint arXiv:2401.01140*.

Khoshkbari, H.; Kaddoum, G.; Selim, B.; Abbasi, O.; and Yanikomeroglu, H. 2025. Distributed Beamforming in Massive MIMO Communication for a Constellation of Airborne Platform Stations. In *ICC 2025 - IEEE International Conference on Communications*, 4383–4388.

Yang, M.; and Zhang, J. 2024. DFedSat: Communication-Efficient and Robust Decentralized Federated Learning for LEO Satellite Constellations. *arXiv preprint arXiv:2407.05850*.

Zhang, C.; et al. 2025. Energy Efficient Trajectory Control and Resource Allocation in Multi-UAV-assisted MEC via Deep Reinforcement Learning. *arXiv preprint arXiv:2508.00261*.